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Measuring walking convenience of Hong Kong resource recycling services: Utilizing a spatially refined model framework of multiple recyclable wastes

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ABSTRACT

Proper recycling leads to environmental benefits. However, there is still a lack of research on the convenience of walking recycling. We introduced the Walking Recycling Capability Index (WRCI) for paper, metal, plastic, and glass, and conducted a case study in Hong Kong. We initially trained machine learning models for municipal solid waste (MSW) generation and non-negative linear regression models for RW proportions, with R^2 values ranging from 0.68 to 0.95. By synthesizing the above models, we developed a stacking model ($R^2 = 0.6731$) to downscale RWs to 0.25 km^2 scale, which served as the demand-side parameter of the Gaussian Two-Step Floating Catchment Area (Ga2SFCA) for calculating the WRCI. By 2025, nearly 70 % of the areas will have a zero or low WRCI in Hong Kong. Our study innovatively provides a high-resolution RWs generation for walking accessibility analysis, offering decision support for enhancing residents' participation in the circular economy.

Synopsis

This study employed ensemble learning methods to predict the spatially refined distribution of four types of recyclable waste in Hong Kong: paper, metal, plastic, and glass, then assessed the convenience for residents to participate in resource recycling based on walking accessibility.

1. Introduction

Globally, municipal solid waste (MSW) management is increasingly critical, with recycling playing a significant role. According to a World Bank report, 2.01 billion tons of MSW are generated annually worldwide, with 38 % comprising recyclable waste (RW) (Kaza et al.). Experiences from various countries have demonstrated that efficient recycling system construction can yield environmental and socioeconomic benefits to society (Conke, 2018). Accessibility, defined as "the potential for interaction and the ability to overcome spatial separation," serves as a core concept in urban planning (Ashik et al., 2024; Kelobonye et al., 2019). In previous studies, accessibility served as a key indicator for evaluating the suitability, social equity, and convenience of public service facilities, such as parks (Yang et al., 2024), schools (Cui et al., 2024), transportation hubs (Cui et al., 2022), and MSW treatment facilities (Zhang and Lin, 2023), thereby providing decision-making support for further enhancing the service capacity. Especially the convenient distances and strategic locations increase public participation in recycling (Valenzuela-Levi and Flores-Castillo, 2023). As a model for MSW recycling in megacities, Hong Kong has established thousands of recycling facilities in recent years that are near residential areas. It has become an important task for the local government to further improve the current recycling efficiency and public participation. Therefore, proposing a robust accessibility evaluation method to enhance local resource recovery efficiency is of great significance.

Walking accessibility should be prioritized in the location of recycling facilities. From a psychological perspective, recycling facilities located at long distances significantly reduce residents' willingness to participate in recycling (Lange et al., 2014). Among various approaches to recycling facilities, residents have identified walking as the most satisfactory mode (Zhuo and Yan, 2022b). Tao et al. demonstrated that walking accessibility to public facilities like markets, restaurants, and parks significantly impacts the likelihood and duration of walking trips (Tao et al., 2023). Pan and Deng improved the accessibility of healthcare allocation using a two-step optimization approach to achieve a more equitable distribution (Pan et al., 2023). Dai simulated residents' travel

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characteristics through the Gaussian Two-Step Floating Catchment Area (Ga2SFCA) method, reflecting the supply-demand relationship between residents and public facilities (Dai, 2010). Currently, Ga2SFCA is widely used in various studies because the S-shaped decay of the Gaussian function conforms with residents' daily mobility patterns, which assumes that mid-range accessibility decays faster than short-range and long-range accessibility. This approach not only accounts for the strong influences in nearby areas but also effectively represents weaker influences in distant areas, enabling a more nuanced and realistic evaluation of accessibility (Zhang et al., 2022b). The Ga2SFCA method demonstrates strong adaptability, but research on its application in walking recycling scenarios remains scarce. The walking accessibility evaluation method based on Ga2SFCA is expected to incorporate residents' walking mode into the decision-making process for optimizing recycling systems, thereby enhancing resident participation.

The research by Milbrandt et al. demonstrated that the development of the recycling industry relies on reliable and high-resolution spatial data (Milbrandt et al., 2022). Therefore, our study introduced spatial distributions of various RWs to improve Ga2SFCA demand-side calculations and analyze the convenience of walking-based recycling. However, there is still a lack of understanding of the RW's distribution at small spatial scales. Madden et al. predicted multiple MSW at the residential property scale but didn't differentiate RW types or consider socioeconomic factors (Madden et al., 2021). Ghinea et al. forecasted the composition of MSW based on socioeconomic conditions, but the spatial distribution of different RWs was not included in their study (Ghinea et al., 2016). At present, paper, plastic, metal, and glass are the four most produced RWs, accounting for 66 % of Hong Kong's total RW mass (HKEPD, 2023), and the local government is working to improve public participation in their recycling system. There is an urgent need for Hong Kong to understand the spatial characteristics of major RWs in the future to rationally optimize the facility layout. However, there remains a lack of RW generation data sufficient to support high-precision walking accessibility analysis. Thus, we chose Hong Kong as the study area to develop a spatiotemporal regression model framework of the four above RWs for supporting the walking accessibility analysis of recycling.

Our research aims to identify the existing deficiencies in residents' walking participation in recycling, which is one of the most basic and important parts of the city's resource recycling system. In this study, machine learning models of MSW and a modified non-negative linear regression model for the proportion of paper, metal, plastic, and glass, were established to understand the recycling potential at the city and district scale. Then, the downscaling ensemble learning method was used to clarify the refined generation of different RWs at a grid scale of 0.25 km^2. Finally, we proposed the Walking Recycling Capability Index (WRCI) based on the spatiotemporal regression of four types of RWs to evaluate the convenience of residents' walking participation in recycling, which fully considers the spatial imbalance between RWs generation and facility distribution. This study combined advanced spatiotemporal amount predictions of four RW types with a novel

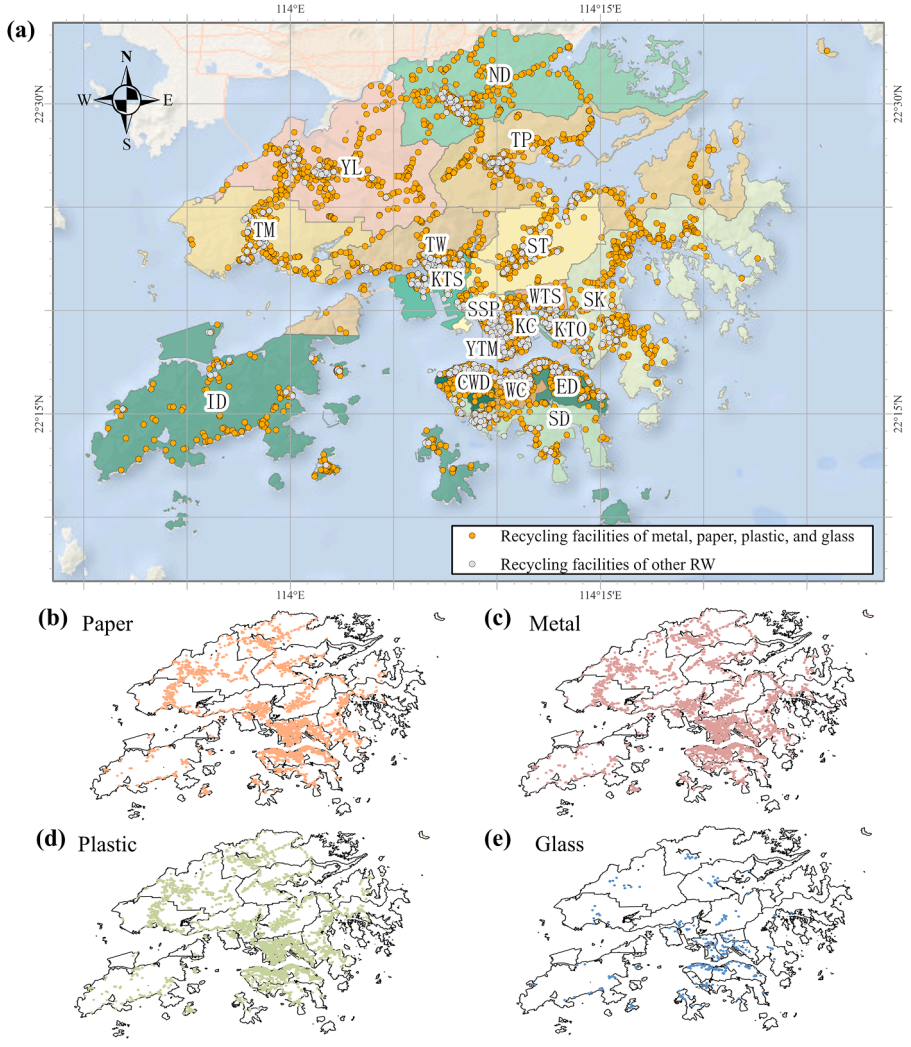


Fig. 1. Study area (a) and recycling facilities distribution of paper (b), metal (c), plastic (d), and glass (e).

walking accessibility evaluation to guide policy for recycling system enhancement.

2. Materials and methods

2.1. Study area

Hong Kong, located in southern China, consists of 18 districts (Detailed in Table S1) and spans an area of 1106 km^2. By the end of 2023, its population had reached 7.47 million, resulting in a population density of 6900 people/km^2 (Fig. 1a).

As an international financial center, Hong Kong generates substantial amounts of MSW. In 2022, the city produced 11 Kt of daily MSW (HKEPD, 2023). Effectively managing and recycling MSW has become a crucial task for Hong Kong. In 2013, the Hong Kong Environmental Bureau published the "Waste Blueprint for Hong Kong 2013–2022", initiating a resource recycling campaign (HKEB, 2013). Since then, Hong Kong has established 3858 convenient recycling facilities (Fig. 1a), such as frontline collectors, preprocessors, and exporters, encouraging residents to voluntarily participate in recycling (Chen et al., 2023; Kua et al., 2024). In Hong Kong, paper, plastic, metal, and glass are the principal categories of RW, and their recycling facilities are distributed throughout Hong Kong, which are 2748, 3035, 3101, and 250 respectively (Fig. 1b–e). However, in 2022, only 25 % of the MSW in Hong Kong was recycled (HKEPD, 2023), indicating significant shortcomings in the recycling system. In the "Waste Blueprint for Hong Kong 2035" published in 2021, Hong Kong further emphasized the completion of recycling infrastructure by 2025 (HKEB, 2021). Our research results will provide policy insights aimed at enhancing the local recycling system's convenience.

2.2. Spatial distribution model of RW

To assess the reasonable layout of recycling facilities, we first constructed a spatial grid prediction model framework of RWs. Our modeling framework integrated three interlinked parts: the base machine learning models of MSW generation, the modified non-negative linear model of four RWs proportion, and the grid downscaling stacking model of RW generation (Fig. 2). The production of metal, glass, paper, and plastic at a 0.25 km^2 (500 m × 500 m) scale for 2025 was predicted using this framework. The future explanatory variables that were used for RWs spatiotemporal prediction in 2025 were obtained by exponential smoothing regression of historical data. Furthermore, the data processing and model building were implemented using Excel, ArcMap 10.7, and Python 3.7.

2.2.1. The base machine learning models of MSW generation

We first established the regression relationship between socioeconomic characteristics, spatial characteristics, and MSW generation through various decision tree algorithms. The algorithms used in our study include XGBoost (XGB), LightGBM (LGB), and CatBoost (CTB), as detailed below:

(1) XGB (Markovic et al., 2022): Enhances the GBDT algorithm with a refined second-order gradient-based loss function, offering high-speed computation, adaptability, and reduced overfitting tendencies;
(2) LGB (Csizmadia et al., 2022): A gradient boosting framework that employs exclusive feature bundling and gradient-based one-sided sampling, optimized for lightweight architecture and computational efficiency;
(3) CTB (Madani et al., 2021): Based on the GBDT algorithm with a greedy approach, it addresses gradient deviation and prediction offset issues, providing short training time and eliminating the need for categorical variable preprocessing.

We collected panel data from each district of Hong Kong during 2012–2022, aiming to establish a spatiotemporal regression relationship between multi-source panel data and MSW generation (Table. S2). Due to the influence of population, economy, and education level on MSW generation characteristics (Duan et al., 2021; Zhang et al., 2022a), the socioeconomic data includes the total population, populations with primary education, junior high school education, high school education, non-degree post-secondary education, degree-level education, and total disposable income. Then we introduced spatial data such as the build-up areas, total nighttime light intensity, and longitude and latitude coordinates to support the next downscaling. These data were sourced from the government (Census and Statistics Department of Hong Kong, 2023), WorldPop (WorldPop and Center for International Earth Science Information Network (CIESIN), 2018), and NPP/VIIRS nighttime light data. The model was trained and tested using data from 2012 to 2021 (25 % of the data was randomly sampled for testing) and then validated with the data of 2022.

2.2.2. The modified non-negative linear model of four RWs' proportion

Hong Kong currently lacks a district-scale accounting system for the

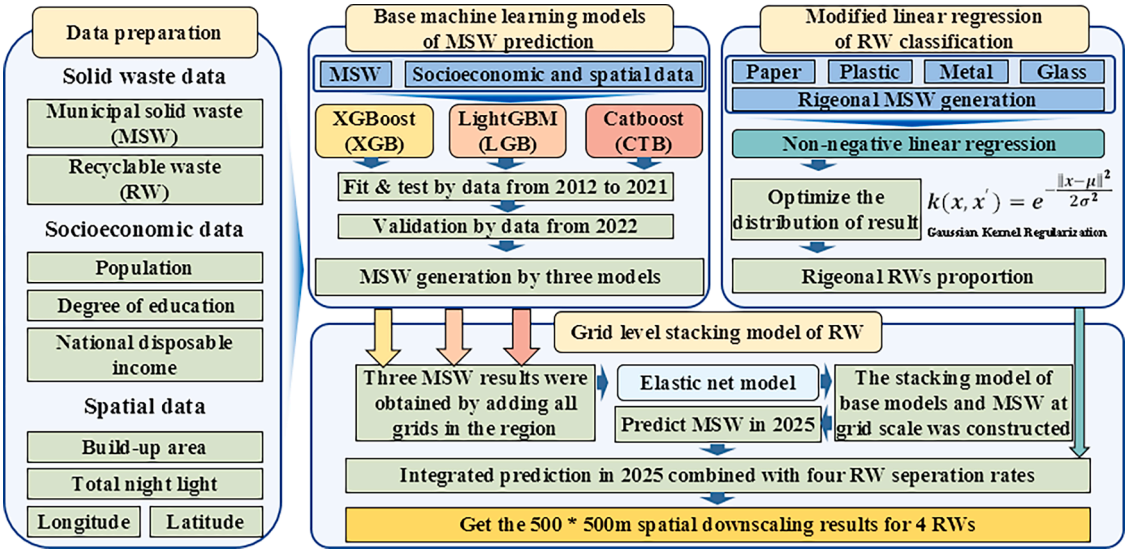


Fig. 2. Model framework of the RW spatial distribution model.

proportion of RW, relying solely on treatment facilities to estimate the whole city's RW proportion. Given the scarcity of data, we employed a non-negative linear regression method, which requires minimal data input. Using the MSW data from each district of Hong Kong from 2012 to 2022 as features (Table 1), four models fitted the spatiotemporal regression relationships between the regional MSW and the masses of paper, metal, plastic, and glass, respectively.

Considering the influence of population structure and economic income on RW components (Gu and Fujiwara, 2009; Li et al., 2011), we assume that the RW distribution characteristics remain relatively consistent within the same region. Kernel functions are widely used to constrain the parameter range for improving prediction accuracy in regression processes (Shi et al., 2022). Further, the Gaussian kernel constraint method, typically employed in model fitting, assumes that each random variable is independent and uniformly distributed within the parameter space (Tian et al., 2024). Previous research has proven that the proportions of various types of RW in China are consistent with the lognormal distribution regularity (Gu et al., 2017). Ideally, distributions such as the beta distribution, specifically designed for proportions bounded between 0 and 1, would be more suitable. However, kernel functions require a distribution model that ensures symmetry and positive-definite properties. This makes it challenging to apply the lognormal or beta distribution for kernel function regression optimization, especially under data limitations. In contrast, the normal distribution proves to be more robust with small sample sizes, and its kernel function is both simple and efficient. Therefore, the method employed in this study is based on constructing a Gaussian kernel derived from the normal distribution. By incorporating the Normal distribution, we smooth the parameters' optimization process and ensure alignment with historical data patterns. The Gaussian Kernel constraint that we introduced during the model fitting process is shown in the following formula:

$$k(x,\mu) = e^{-\frac{x-\mu^2}{2\sigma^2}} \tag{1}$$

where x is the proportion of different RWs in each year; μ is the mean value of the historical proportion; σ is the standard deviation of the data distribution, a measure of dispersion. The Gaussian kernel function $k(x,\mu)$ was incorporated as a regularization term into our non-negative linear model, defining the prior distribution of the solution. Finally, the linear regression coefficients obtained by solving the model were regarded as the proportions of different types of RWs in each district.

2.2.3. The grid downscaling stacking model of RW generation

To clarify the refined spatial distribution of RWs in 2025, we first divided the study area into grids of 500 m × 500 m. Then the trained base models of MSW generation in Section 2.2.1 were independently applied to each grid to estimate the MSW. However, directly extrapolating machine learning models from the district scale to the grid scale would result in significant errors (Gu et al., 2017). Previous research by Wei et al. developed a novel method that ensembles spatiotemporal weighted into machine learning models, achieving simulation of daily 1 km^2 surface NO_2 concentrations under different seasons (Wei et al., 2022). This demonstrates the combination of spatiotemporal weights, machine learning, and ensemble learning methods are promising for optimizing the performance of prediction of multiple RW at fine spatial scale. Therefore, we further employed an elastic net stacking model for ensemble learning. The elastic net method excels in stacking due to its ability to balance bias and variance, handle multi-collinearity, and perform automatic feature selection. By combining L1 and L2 regularization, it ensures robust and interpretable model outcomes (Qayyum et al., 2023). The results of the base models were summed by district and input into the elastic net model to fit the regression relationship with the historical MSW generation in each district. After training, the stacking model can downscale MSW generation through the results of three base models. The explanatory variables that were used for RWs spatiotemporal prediction in 2025 were obtained by exponential smoothing regression of historical data. Finally, we input the three prediction results of each grid in 2025 obtained by the base models into the trained stacking model to obtain the MSW generation at the grid level. According to the district to which different grids belong, by multiplying the corresponding RWs' proportion, we obtained the future spatial characteristics of four RWs in 0.25 km^2 grids.

Table 1
Historical proportion of RWs.

	Paper	Plastic	Metal	Glass
2012	20.53 %	19.68 %	2.58 %	3.11 %
2013	19.09 %	19.54 %	1.85 %	3.70 %
2014	19.65 %	20.60 %	2.14 %	2.91 %
2015	22.22 %	21.49 %	2.32 %	3.61 %
2016	21.69 %	20.61 %	2.34 %	3.18 %
2017	23.60 %	19.79 %	2.35 %	2.71 %
2018	23.65 %	20.51 %	2.44 %	2.63 %
2019	24.46 %	20.98 %	2.32 %	2.29 %
2020	24.46 %	21.40 %	2.20 %	1.69 %
2021	19.67 %	20.52 %	2.30 %	1.94 %
2022	20.17 %	21.29 %	2.23 %	1.90 %

2.3. Walking accessibility assessment of RWs recycling

Walking accessibility is assessed by measuring the walking distance people can achieve within a given time frame. Aiming to evaluate the accessibility of recycling facilities to residential areas, we proposed the WRCI, which is modified based on the Ga2SFCA method. The Ga2SFCA method has been widely used in public service assessment demonstrating the accessibility of different travel modes such as walking, motor vehicle, and non-motor vehicle (Dai, 2010; Liang et al., 2023; Zhang et al., 2022b). The WRCI method based on the Ga2SFCA approach fully considers the spatiotemporal imbalance of different types of RWs generated by residents under the influence of factors such as income, education level, and urban development. Compared to traditional studies that focus solely on the geographic distance between residents' residences and facilities, it emphasizes the interaction between walking preferences, facility distribution, and environmental impacts, providing a better reflection of the differences in recycling accessibility across different resident groups. Here, we changed the demand parameters of this method to the spatial refined RWs generation. The revised walking accessibility for RWs is the WRCI, and the formula is as follows:

$$\beta_i = \sum_{j,j\in\left[d_{ij}\le d0\right]} \frac{G_j f(d_{ij})}{\sum_{k,k\in\left[d_{jk}\le d_0\right]} P_k f(d_{kj})} \tag{2}$$

The formula for the Gaussian distance decay function $f(d_{ij})$ is as follows:

$$f(d_{ij}) = \begin{cases} \dfrac{e^{-\frac{1}{2}\left(\frac{d_{ij}}{d_0}\right)^2} - e^{-\frac{1}{2}}}{1 - e^{-\frac{1}{2}}} & \text{if} d_{ij} \le d_0 \\ 0 \text{ if} d_{ij} > d_0 \end{cases} \tag{3}$$

where the β_i indicates the WRCI of grid i; d_0 is the maximum effective range threshold, since 30 min are often considered the farthest threshold for walking in previous studies (Sharma and Patil, 2024), 2500 m was set in this study as the farthest effective radius of WRCI; G_j is the service ability of recycling facility j, based on Chen et al. 's field statistics at multiple recycling facilities in Hong Kong (Chen et al., 2023), the official Green@Community system of Hong Kong recovered the median of wastepaper collection is at most can reach 0.7 Kt/ year (Chen and Steuer, 2024), and due to data limitations, we standardized the service

capacity of all RW facilities to 1 kt/yr. While this simplification may not fully align with reality, it reduced computational complexity and enabled parallel comparisons across different facilities and types of RWs; P_k is the RWs generation of each grid within the effective range d_0 of residential area k, Kt/Yr; d_{ij} is the distance between the grid i and the recycling facility j; d_{kj} is the distance between the recycling facility j and residential area k within its effective range, and d_0 is the maximum distance threshold. The functions $f(d_{ij})$ and $f(d_{kj})$ provide the contribution weights for different distances to the accessibility.

Further, given that 800–1200 m is often considered an ideal distance for walking (Shao et al., 2022; Zhu et al., 2024), we used the presence of at least one recycling facility within 1000 m, which can meet the average recycling demand of the whole study area (the mean value of the summed P_k), as a benchmark to evaluate the WRCI. When the WRCI of a certain RW in a grid is less than the benchmark, it can be considered that the RW has relatively low walking accessibility (Low-WRCI areas), so the convenience of recycling needs to be strengthened. In addition, it is worth noting that when the WRCI of a grid is equal to 0 (Zero-WRCI areas), it proves that the area is difficult to recycle by walking, and there is an urgent need to build additional recycling facilities in the surrounding area. However, even if the WRCI is much higher than the benchmark (High-WRCI areas), it does not prove that the recycling system is perfect, and it is necessary to relocate recycling facilities in high-WRCI areas to low-recyclable areas to achieve a spatial balance in the capacity of walking recycling.

3. Results

3.1. MSW generation and RWs proportion in 2025

The three base models, XGB, LGB, and CTB, were used for the regression analysis of MSW generation, each yielding relatively accurate results. It is worth noting that there are differences in the models' test and validation performance (Table 2). Among them, XGB exhibited the lowest MAE (43.10) and MAPE (7.77 %) and the highest R^2 value of 0.9075 in test performance, indicating the highest test accuracy. However, in the validation set, LGB showed an MAE of 71.44, MAPE of 13.32 %, and R^2 value of 0.9060, slightly outperforming XGB. Combining the comparison of the three models with actual data (Fig. 3a), LGB demonstrated more stable performance in both the test and validation sets, indicating stronger generalization ability. The LGB performed best for MSW prediction. Although CTB had relatively lower accuracy, all three models exhibited accuracy in evaluating MSW generation. Chung et.al constructed a multivariate autoregressive model of Hong Kong to simulate the MSW from 2011 to 2036, however, the percentage error of their study with the real MSW generation was 19.3 % (Chung, 2010), which is higher than our study (1.7 %). Then our models were used to predict the MSW in Hong Kong until 2025. The LGB model projected a 3.7 % increase in MSW by 2025 compared to 2022, reaching 4212.18 Kt. In contrast, the XGB and CTB models forecasted a slight decrease in MSW generation, with reductions of 1.1 % and 2.8 % compared to 2022, respectively (Fig. 3b). In summary, based on the average results of the three models, the generation of MSW in Hong Kong is expected to be 4061 Kt/Yr by 2025.

To develop a Gaussian kernel-enhanced non-negative linear regression model for predicting the RW proportions, we analyzed the statistical characteristics of historical RW proportions in MSW (Fig. 3c). In the annual MSW, paper, and plastic account for more, reaching an average of more than 20 %, while metal and glass average about 2–3 %. Then we used the annual MSW generation of each district as the explanatory variable and established Gaussian kernel and non-negative linear regression models for the four types of RWs, which were consistent with the data distribution shown in Table 3.

The linear regression coefficient obtained was the RWs proportion of each district (Fig. 3d). The RWs proportion models' R^2 of paper, metal, plastic, and glass were 0.9336, 0.7716, 0.9550, and 0.6852, respectively. In previous studies, Ma et al. used the back-propagation neural network to predict building waste, plastic, fabric, and wood proportion in China's MSW, whose R^2 was only 0.26–0.29, and the model construction process used explanatory variables such as population and economy, which was more complex than our RWs' proportion models (Ma et al., 2020).

In summary, Hong Kong will have a total RW proportion of 48.5 % in 2025, and its various districts will fluctuate roughly between 38.2 % and 64.7 %. Paper, metal, plastic, and glass will generate an average of 2590.12, 255.36, 2324.06, and 232.28 t/day in 2025, respectively, indicating that Hong Kong's MSW has significant recycling potential. Among the RWs, paper and plastic will account for the vast majority. Eastern areas such as KTO and SK areas will generate more plastic, both reaching 22.5 % in MSW. Southeastern districts such as CWD, TM, TW, ST, and ID areas will produce more paper up to 35 % of MSW. Glass will generate more in southern districts such as WC and SD areas, exceeding 4.5 %. Metal will remain relatively stable in most districts, accounting for an average of 2.2 %. Realistically speaking, high-income areas such as WC (Census and Statistics Department of Hong Kong, 2023) have more active commercial activity, bringing in more plastic and glass waste generation (Babayemi et al., 2019). The proportion of the four RWs in our study not only resembles the characteristics of Hong Kong over the past decade (Fig. S3) (HKEPD, 2023) but also shares features with developed countries like Switzerland and Germany, which have comparable economic levels with Hong Kong (Gu et al., 2017).

3.2. Spatial downscaling distribution of RWs generation

In this study, we constructed a stacking model for evaluating RWs at the 500 m × 500 m grid scale through a stacking model, then validated it with data from 2022 (Fig. 4a, b). For intuitive comparison, it can be found that in 2022, the spatial density of MSW generation calculated based on the real data (Fig. 4a) is consistent with the spatial density of MSW generation after downscaling (Fig. 4b). The stacking model constructed by using elastic net method obtained an accurate result (R^2_{test} = 0.7637), and areas with high levels of MSW density are concentrated in the Kowloon Peninsula, such as YTM, KT, SSP areas, etc. In previous studies, Li and Blomberg et al. integrated multiple machine learning algorithms on a scale of ZIP Code Tabulation Area based on the NN-GTWR algorithm with an R^2 of 0.63 (Li et al., 2021); Li et al. employed the Kuznets curve to downscale the provincial and municipal industrial waste to the county scale in Dongguan, which is adjacent to Hong Kong, they achieved an R^2 of 0.8000, which was close to our study (Li et al., 2022). However, it is worth noting that our study achieved a great performance under a more refined spatial resolution of 0.25 km^2. Next, our stacking model was validated by 2022 data ($R^2_{validation}$ = 0.6731). The model framework still showed stable fitting accuracy in the data that was not involved in the training set, which indicated that our method has the potential to realize accurate predictions for the future. We combined the results of the downscale model with the proportion prediction model for RW, resulting in a spatial distribution for the four types of RWs, as shown in Fig. 4c–f. Due to Hong Kong's extensive mountainous areas, oceans, and islands, not all spatial grids generate RW. Out of 5518 grids, only 2333 produce RW more than 1 t/year, accounting for about 42.27 %. In 2025, the maximum amount of paper, metal, plastic, and glass produced inside each grid is expected to be 1669, 110, 980, and 493 t/year, respectively. The grids with high RW

Table 2
Performance of models.

Model	Test performance			Validation performance		
	MAE	MAPE	R^2	MAE	MAPE	R^2
XGB	43.10	7.77 %	0.9075	75.80	13.48 %	0.8834
LGB	49.15	8.97 %	0.8892	71.44	13.32 %	0.9060
CTB	54.41	10.03 %	0.8774	82.80	14.64 %	0.8694

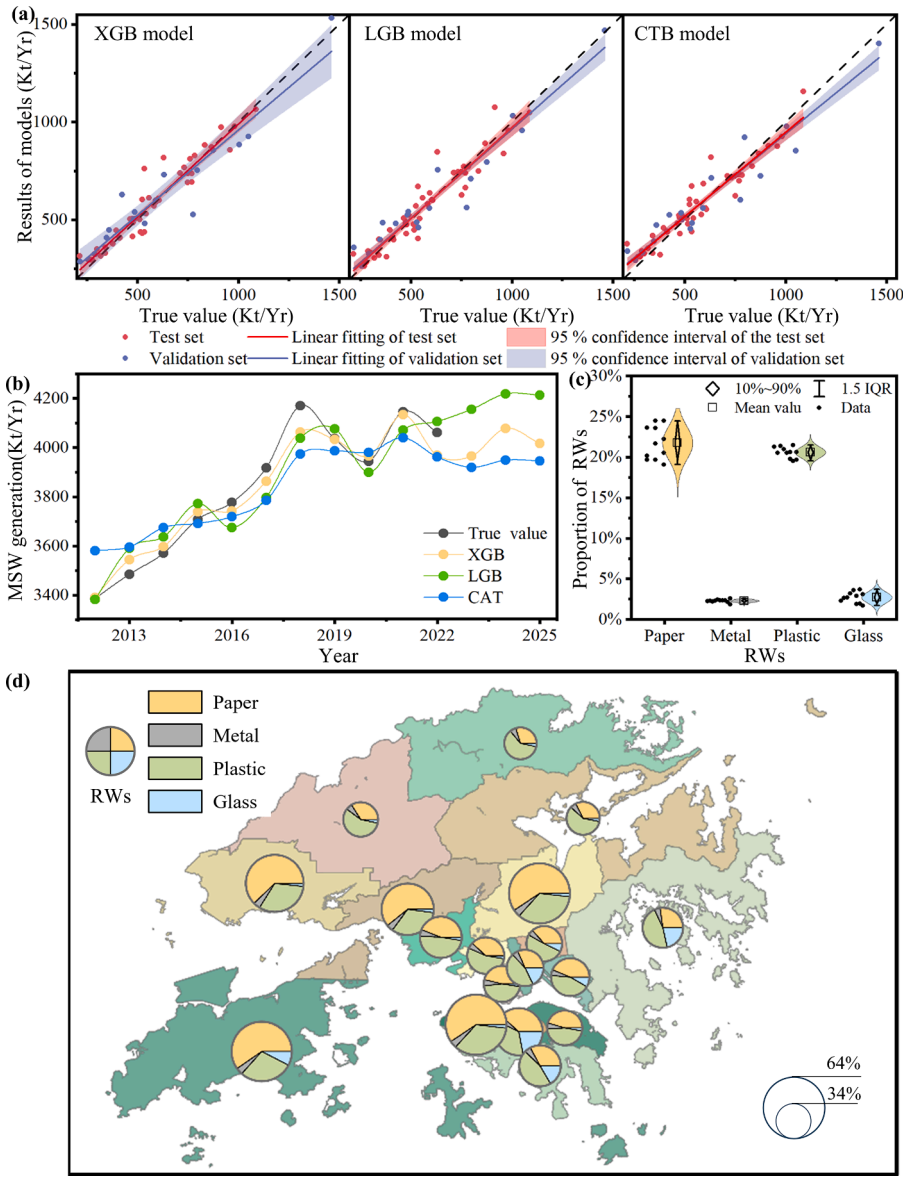


Fig. 3. MSW generation and RWs proportion characteristics in Hong Kong. (a) Comparison of the results of the three base models with the actual data. (b) The true values and prediction values of MSW generation in 2012–2025. (c) Proportion of RWs in Hong Kong from 2012 to 2022. (d) The proportion of RWs in each district in 2025.

Table 3
Statistical characteristics of RWs.

	Paper	Plastic	Metal	Glass
μ	21.74 %	20.58 %	2.28 %	2.70 %
σ	0.0194	0.0065	0.0017	0.0065

production will be primarily concentrated in the Kowloon Peninsula and Hong Kong Island, which are also the most populous and economically vibrant districts in Hong Kong. The downscale results were used to express the RWs' recycling demand in the WRCI analysis.

3.3. WRCI assessment of recycling

Using the Ga2SFCA method, we calculated the WRCI for each spatial grid in Hong Kong that generates RWs. Firstly, we took the center of 0.25 km^2 grids in Hong Kong as the starting point, searched for all recycling facilities within a radius of 2.5 km (Fig. 1b–e), and calculated the WRCI for each type of RWs (Fig. 5a–d). Overall, the average WRCI of each grid in Hong Kong will be as follows in 2025: metal (1.33), glass (0.36), plastic (0.15), and paper (0.14). Given that at least one recycling facility within 1 km is a benchmark, the WRCI benchmarks for paper, metal, plastic, and glass will be 0.06, 0.62, 0.34, and 0.06, respectively. Except for the paper, the average WRCI of RWs will reach the benchmark. However, at the single grid scale, 10.95 %, 9.83 %, 9.83 %, and 35.98 % of grids will be zero-WRCI areas, respectively, and 59.93 %, 62.09 %, 62.56 %, and 34.74 % of grids will be low-WRCI areas, respectively. In summary, about 70 % of the area of all four types of RWs will still need to be improved for walking recycling, and the convenience of glass recycling has the largest service blind spot.

In Fig 5a–d, we selected six representative regions to illustrate typical results that WRCI assessment might demonstrate (Fig. 5e). Region 1 will have a WRCI of 0, primarily due to the lack of relevant recycling facilities in the vicinity, causing walking-based RW recycling infeasible; Both Regions 2 and 3 will exhibit relatively low WRCI but for different reasons: Region 2 will generate less RW compared to Region 3, with facilities sparsely distributed; in contrast, Region 3 will have the

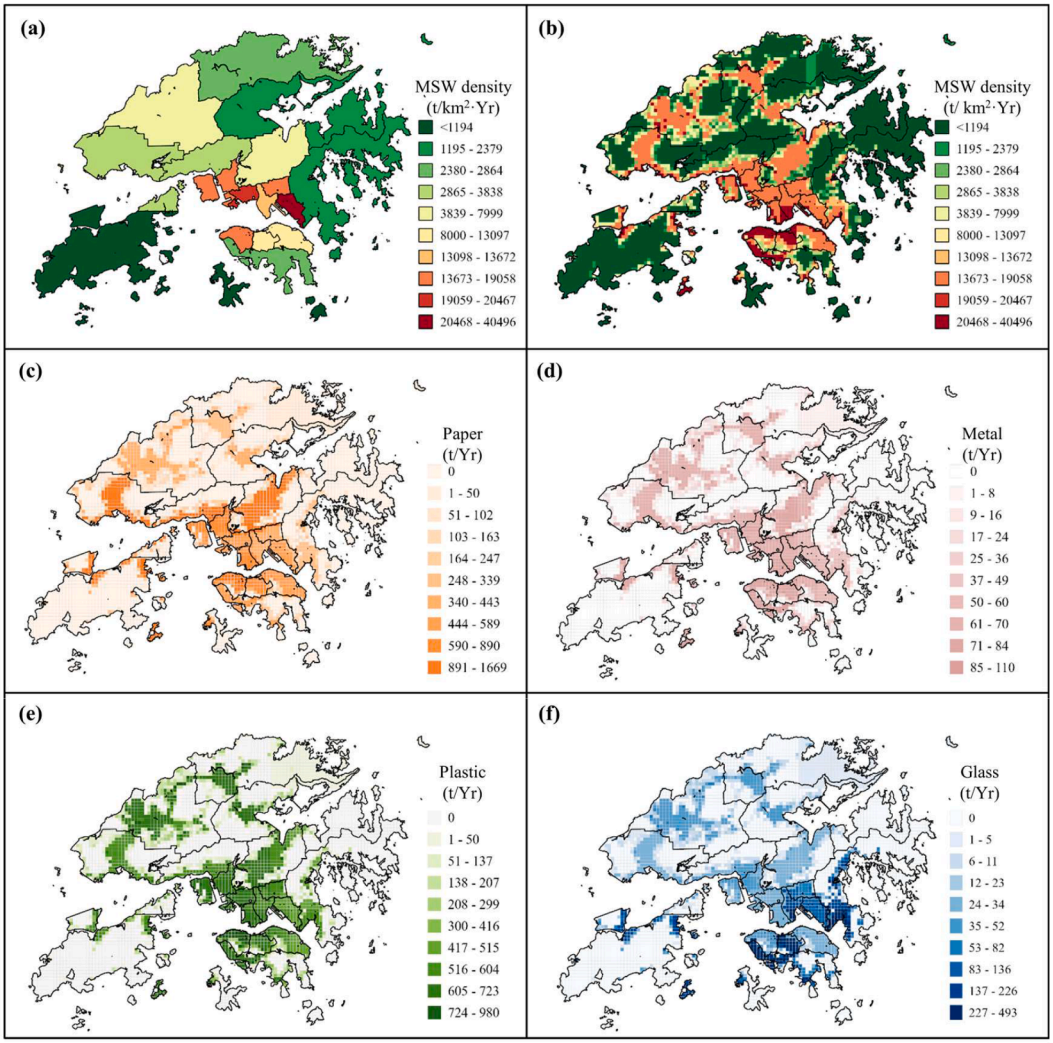


Fig. 4. Spatial distribution of MSW and RWs. (a) Spatial density of MSW in 2022. (b) Downscaling prediction results of MSW in 2022. (c) Spatial distribution of paper in 2025. (d) Spatial distribution of metal in 2025. (e) Spatial distribution of plastic in 2025. (f) Spatial distribution of glass in 2025.

highest RW production density in Hong Kong, whose recycling facilities was unable to meet its high recycling demand. In Region 4, a lot of grids will be marked in yellow, indicating that the facilities in this area are relatively moderate and will satisfy the benchmark of RW recycling. Regions 5 and 6 will show relatively high WRCI values but differ in their characteristics: Region 5 has a low RW generation and few recycling facilities, whereas Region 6 has both high RW generation and a greater number of facilities. Nonetheless, residents in both regions demonstrate strong walking recycling capabilities.

There is a prominent spatial imbalance in the WRCI. We further calculated the WRCI in 18 districts of Hong Kong respectively (Fig. 5f). First we found that due to the similarity of the distribution, the WRCI of paper, plastic, and metal will show similar spatial characteristics. There are a total of 9 districts, including KTO and KTS are all low-WRCI areas for the three RWs. Furthermore, except for paper recycling in the SK area, high-WRCI areas will be less than 50 % simultaneously. It can be seen that not the distribution span of facilities, but the total service capacity of facilities that will limit WRCI, and the mitigation may be achieved by increasing the distribution density of facilities; In addition, despite the minimal number of glass recycling facilities, they will achieve the most of areas (29.28 %) with a high-WRCI. Meanwhile, 35.98 % of grids cannot recycle glass by walking, despite similar production levels of glass and metal, the scarcity of glass recycling facilities calls for an improvement in their numbers. In summary, the current recycling facilities in Hong Kong will be incompatible with the walking willingness of residents to participate in future RW recycling.

We should consider utilizing the WRCI assessment method as an important tool for urban decision-making. The spatial positioning of recycling facilities relative to the living areas is often considered a crucial component in convenient recycling systems (Zhuo and Yan, 2022a). It is necessary to increase the service capacity of recycling facilities in the low-WRCI areas, moreover, existing zero-WRCI areas should be covered by additional recycling facilities, especially in densely populated areas in the middle of Hong Kong such as the Kowloon Peninsula and Hong Kong Island. However, the high-WRCI area does not imply a sound recycling system, but also a relative glut of service capacity, which is most evident in the recycling of metals. As the previous study indicated, unofficial recycling has always been an important part of the current Hong Kong recycling system, which is incoherent and profit-oriented (Chen and Steuer, 2024). It is crucial to either add new recycling facilities or relocate existing ones from high WRCI areas to low or zero WRCI areas, shifting the focus from profit-oriented to convenience-oriented operations. By strategically relocating and adding facilities with WRCI assessment, we can ensure a coherent spatial distribution of recycling capabilities by walking, which could enhance public participation and effectively increase Hong Kong's recycling rates.

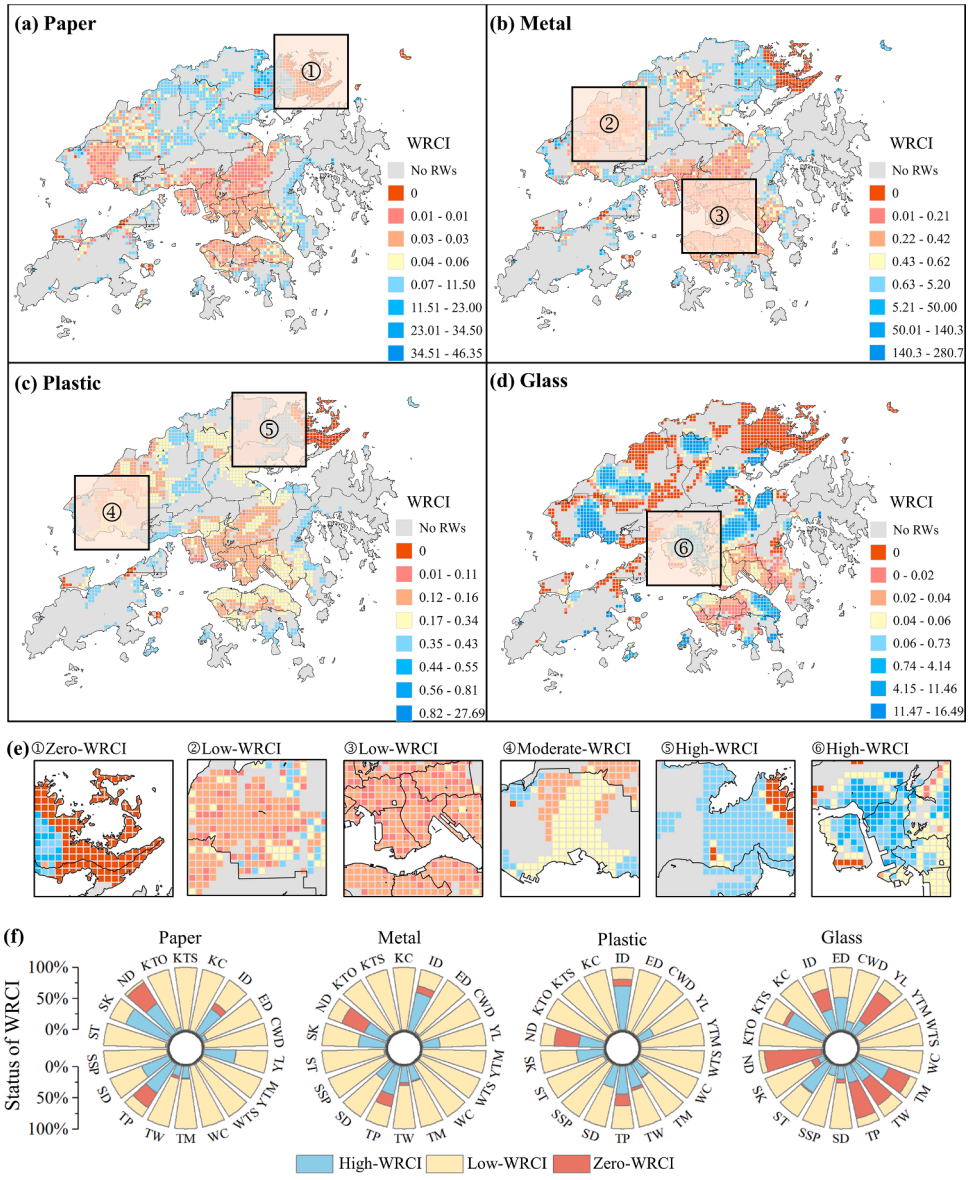


Fig. 5. WRCI of Hong Kong in 2025. (a) WRCI of paper. (b) WRCI of metal. (c) WRCI of plastic. (d) WRCI of paper. (e) Example WRCI under different conditions. (f) WRCI status of 18 districts in Hong Kong.

4. Discussion

This study initially constructed a spatiotemporal forecasting framework of paper, metal, plastic, and glass, which employed ensemble learning to mitigate the precision loss typically incurred after downscaling in a 0.25 km^2 grid scale (Li et al., 2024). The research elucidated the significance of various characteristics in RWs distribution by utilizing the SHapley Additive exPlanations (SHAP) methodology. We calculated the contributions of explanatory variables in each grid for the year 2025 (Table. S1 and Fig. S2). Previous studies pointed out that education development influences the generation patterns of paper and plastic waste (Gu and Fujiwara, 2009; Li et al., 2011). Notably, the population with a junior high school education had the highest mean SHAP values of all types of RWs, indicating it as the most prominent factor. It is noteworthy that for paper and plastic waste, the high school population also emerged as a significant factor, suggesting that education primarily impacts the spatiotemporal characteristics of RWs, which aligns with Nesli et al.'s finding that lower-educated populations have a more pronounced impact on recycling behavior (Sena Sağlam and Aydın, 2024). The total population remains in the top five of characteristic importance for the four RWs, similar to the phenomenon in the MSW predictions (Duan et al., 2021; Zhang et al., 2022a). Furthermore, the influence of latitude and longitude coordinates on RW generation is also significant, indicating cluster effects and spatial heterogeneity of RW generation, with higher RW output near Victoria Harbour. Gu et al. conducted a Monte Carlo analysis of MSW across China, revealing spatial variations of waste components associated with economic levels, including food waste, ash, paper, plastics, and other waste (Gu et al., 2017). In our study, different RWs exhibited varying degrees of dependence on income too, especially glass showing the highest dependency. To further clarify the impact, we calculated the average contribution of each feature to RW production (Fig. S2). On average, all features promoted the RW growth generally. The effects of population with middle and high school education, geographic coordinates, and income are the five most pronounced factors. A previous study found a gradual increase in the proportion of RWs across regions in China. This trend has been attributed to the accumulation of education and wealth, which drives individuals toward more diverse needs and results in increasingly complex discarded materials (Gu et al., 2015). However, different features exhibited varying SHAP values

across grids, meaning that the contribution of each feature is nonlinear and spatially heterogeneous. Moreover, the validation of the RWs' proportion prediction also demonstrated the effectiveness of feature selection, whose R^2 of paper, metal, plastic, and glass were 0.9336, 0.7716, 0.9550, and 0.6852, respectively. Considering these factors collectively to further improve the recycling system is of significant practical importance.

Based on the high-precision spatiotemporal characteristics of RWs, the WRCI demonstrated multifaceted application value in the enhancement of recycling systems. Previous studies integrated socioeconomic differences and population lifestyles into the accessibility assessment process. For instance, Kulkarni et al. conducted a travel survey with individual-level travel perception data as income—and predicted network-level walking perception for different socioeconomic groups (Kar et al., 2024). Similarly, Elias et al. assigned different walking speeds for various demographic age groups, and then assessed accessibility at different time thresholds (Willberg et al., 2023). This study utilized a variety of explanatory variables during the RW prediction phase, providing a high-precision demand-side data foundation for WRCI's accessibility assessments, and supporting WRCI's comprehensive consideration of education, population, income, urban construction, and spatial location, which is crucial for optimizing collection routes, improving waste management efficiency, and enhancing the overall effectiveness of recycling systems. Moreover, fine-scale, more complex kinds, and longer time-span of RW generation predictions, as well as the walking accessibility analysis that considers the travel behaviors of different people, and real-time traffic networks, may be the future development directions of the WRCI method. Additionally, WRCI can support the future design of economic incentive programs effectively. Hong Kong motivates residents to recycle by assigning GREEN$ balance and corresponding gift rewards to various types of RWs (Chan et al., 2024). Dongying, China (Sun and Asari, 2024) and Seoul, South Korea (Hong et al., 2014) also improved RW recycling efficiency through well-designed economic incentives. Not only will the prediction and recycling capacity evaluation of RWs based on the spatial refinement of WRCI serve as a basis for formulating regional economic incentive policies, but economic incentives also have the potential to extend residents' recycling distance thresholds (Chan et al., 2024). Integrating WRCI with economic incentives represents an important direction for future research.

5. Conclusion

This study proposed the concept of WRCI to assess the walking convenience of recycling with a spatial downscaling model framework of RWs. The methodology was applied to Hong Kong, identifying areas where the walking recycling capacity will be insufficient by 2025, the following conclusions can be drawn:

1. Our methods maximized the use of existing data by separately modeling the quality and composition of multiple RWs, yielding accurate testing and verification results. Among them, the three machine learning models of MSW (R^2 values ranging from 0.86 to 0.90) utilized socioeconomic data and spatial data, which not only ensured that the MSW predictions were consistent with societal characteristics but also guaranteed the potential for spatial downscaling. On the other hand, the non-negative linear regression models (R^2 values ranging from 0.68 to 0.95) required less data and refined outcomes within a credible range by the Gaussian kernel constraint. We projected that Hong Kong will generate 4150 Kt of MSW, with paper, metal, plastic, and glass accounting for 23.28 %, 2.29 %, 20.89 %, and 2.09 % by 2025, respectively. The proposed methodological framework merits application in the further application of solid waste management.
2. We proposed an innovative downscaling framework of RWs that integrates various models through ensemble learning methods. This integration encompasses the mass of MSW and the spatial proportion disparities of four RWs. Hence, we predicted the distribution of RWs in the 5518 grid cells of 0.25 km^2 in Hong Kong by applying an elastic net model ($R^2_{test} = 0.7637$, $R^2_{validation} = 0.6731$). The introduction of explanatory variables with spatial attributes and the application of the stacking model enable the accuracy of predictions of RWs at a small grid scale. The downscaled model predicts that in 2025, 42.27 % of grids in Hong Kong will generate over 1 t/year of MSW, with high RW concentrations mainly in Kowloon and Hong Kong Island. The results fully identified the spatial unbalance of four RWs and supported the analysis of walking recycling capability.
3. The WRCI aims to assess the walking accessibility of recycling facilities and highlight their spatial imbalance. We established a benchmark requiring at least one recycling facility within 1 km to compare the WRCI for the four RWs: metal (1.33), glass (0.36), plastic (0.15), and paper (0.14). We identified zero-WRCI areas (grids with no accessible recycling within walking distance) and low-WRCI areas (WRCI lower than the benchmark). The glass has the weakest walking recycling capability, with the most zero-WRCI areas comprising 35.98 % of grids. We found that the walking recycling accessibility of all 18 districts in Hong Kong where will be challenging in 2025, and recommend increasing or relocating facilities according to the results of the WRCI.

In summary, our calculation of WRCI fully considered the uneven spatial distribution of RWs generation and recycling facility. Governments could integrate our downscaling method and WRCI assessment to better understand and enhance residents' participation in recycling. By referencing the WRCI, we could optimize the service capacity of recycling facilities to strategically enhance the recycling rates of RWs, thereby boosting economic and environmental benefits. In addition, our approach can flexibly adapt to the adjustment of key parameters such as distance thresholds, travel mode, and waste type. We are expected to further improve our framework to evaluate the recycling accessibility of multiple solid wastes in other cities around the world under different needs.

CRediT authorship contribution statement

Tianrui Zhao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Xuanlong Shang:** Writing – review & editing, Writing – original draft. **Lipin Li:** Methodology, Data curation. **Weijia Li:** Supervision. **Yanliang Li:** Software. **Ming Guan:** Software. **An Ding:** Supervision, Investigation, Formal analysis, Conceptualization. **Jun Zhang:** Writing – review & editing, Supervision, Formal analysis. **Yu Tian:** Supervision, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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